

Problem 1:

Base case: $T(2) = 2 = 2\log_2(2)$

Assume:

$$T(n/2) = (n/2)\log_2(n/2)$$

1. $T(n) = 2T(n/2) + n$
2. $T(n) = 2((n/2)\log_2(n/2)) + n$
3. $T(n) = n\log_2(n/2) + n$
4. $T(n) = n(\log_2(n) - 1) + n$
5. $T(n) = n\log_2(n)$

->proven.

Problem 2:

To sort $A[1\dots n]$, we recursively sort $A[1\dots n-1]$ then insert $A[n]$ into the sorted array

$$T(n) = T(n-1) + A[n]$$

To insert, we compare $A[n]$ with elements in $A[1\dots n-1]$ from $n-1$, if the value of $A[n]$ is smaller than all other numbers in the array, then it takes $T(n-1)$ comparisons to find its position.

Base case: $T(1) = 0$ since there's only 1 comparison and it takes $n-1$ comparisons
 $\Rightarrow 1-1 = 0$

Worst case running time: $T(n) = T(n-1) + (n-1)$ ($n > 1$, $T(1) = 0$)

Problem 3:

Pseudocode:

```

sortingNegativePositive(arr) {
  left = 0; //the begin of the array
  right = size of arr - 1; //the end of the array
  while (left < right) { //stop when left meets right
    if (arr[left] < 0):
      left = left + 1; //move to the right
    else if (arr[right] > 0):
      right = right - 1; //move to the left
    else:
      swap arr[left] to arr[right];
      left = left + 1;
      right = right - 1;
  }
}

```

```
};  
}
```

Problem 4:

-In a binary search tree, each element can possibly have maximum two children

$\Rightarrow n = 2^k - 1$ nodes total when we have a perfectly balanced tree

$\Rightarrow 2^k = n + 1$

$\Rightarrow k = \log_2(n+1)$

-The number of nodes at level i only is $2^{(i-1)}$

$S_n = \sum_{i=1}^k i \times 2^{(i-1)}$

$S_n = (k - 1)2^k + 1$

$S_n = (k - 1)(n + 1) + 1$

$S_n = (\log_2(n + 1) - 1)(n + 1) + 1$

$S_n = (n + 1)\log_2(n + 1) - n - 1 + 1$

$S_n = (n + 1)\log_2(n + 1) - n$

Average = $S_n/n = ((n + 1)\log_2(n + 1) - n) / n$

$A_n = \log_2(n + 1) - 1$

Base case is 1

Worst case is $\text{floor } \log_2(n) + 1$

Average case = $\log_2(n) - 1$